Functional Annotation Report: fruK (PP_0794 / Q88PQ4) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

fruK (locus PP_0794; UniProt **Q88PQ4**) of *Pseudomonas putida* KT2440 encodes **1-phosphofructokinase (fructose-1-phosphate kinase, FruK; EC 2.7.1.56)**, a soluble cytoplasmic enzyme of the **PfkB (ribokinase-like) carbohydrate-kinase family**. Its primary function is to catalyze the **ATP-dependent phosphorylation of β-D-fructose 1-phosphate to β-D-fructose 1,6-bisphosphate**:

> β-D-fructose 1-phosphate + ATP → β-D-fructose 1,6-bisphosphate + ADP + H^+ (Rhea:14213)

FruK is the committed cytoplasmic step of the **fructose phosphotransferase (PTS) utilization pathway**. Fructose is imported and simultaneously phosphorylated to **fructose-1-phosphate (F1P)** by the fructose-specific PTS (FruB/FruA); FruK then converts F1P to fructose-1,6-bisphosphate, which is cleaved by fructose-bisphosphate aldolase and channelled into central carbon metabolism. Because *P. putida* KT2440 **lacks glycolytic 6-phosphofructokinase and metabolizes sugars exclusively via the Entner–Doudoroff pathway**, FruK is the obligatory, dedicated kinase that admits fructose to central metabolism. FruK is encoded in the **fruBKA operon**, which is repressed by the regulator **Cra (FruR)** and de-repressed specifically by FruK's own substrate, **F1P** — placing FruK at the heart of a feed-forward regulatory circuit for fructose catabolism.

2. Gene / Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88PQ4	UniProt
Gene name / locus	*fruK* / PP_0794	UniProt (EMBL AAN66419.1)
Organism	*Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125)	UniProt
Protein	Phosphofructokinase → **1-phosphofructokinase**	UniProt
EC number	**2.7.1.56**	UniProt / Rhea
Length	315 aa	UniProt
Family	Carbohydrate kinase **PfkB** family	UniProt SIMILARITY
Domains	PF00294 (PfkB); IPR011611 (PfkB domain); IPR029056 (ribokinase-like); IPR017583 (tagatose/fructose PfkB-type kinase); IPR002173 (PfkB conserved site)	InterPro/Pfam
KEGG	ppu:PP_0794	UniProt xref

Verification outcome — IDENTITY CONFIRMED. The gene symbol *fruK*, the organism, the UniProt catalytic annotation (F1P + ATP → FBP), the EC number 2.7.1.56, and the PfkB/ribokinase-like domain architecture are all mutually consistent and match the literature for 1-phosphofructokinase. A primary study (Yoon et al. 2021, P 33476373) explicitly names **"Pseudomonas putida … the 1-phosphofructokinase FruK"** within the fru operon, giving direct, organism-specific corroboration. There is no evidence of a mistaken-identity conflict with a same-symbol gene in another organism.

> ⚠ Note on nomenclature: In *P. putida* the operon is written **fruBKA** (FruB = PTS EI/FPr-like fusion component, FruK = 1-PFK, FruA = PTS EII permease). In *E. coli* the equivalent operon is **fruFKA**. "FruK" should not be confused with the glycolytic **PfkA/PfkB 6-phosphofructokinases**, which act on fructose-**6**-phosphate and belong (for PfkA) to a different family.

2.1 Genomic context (fruBKA operon) — independent bioinformatic verification

Locus-name resolution against *P. putida* KT2440 (taxid 160488) places *fruK* within a compact, functionally coherent gene cluster, exactly matching the operon architecture in the literature:

Locus	Gene	UniProt	Product
PP_0792	*cra* (fruR)	Q88PQ6	Catabolite repressor-activator (Cra), DNA-binding dual regulator
PP_0793	*fruB*	Q88PQ5	Phosphoenolpyruvate–protein phosphotransferase (PTS multiphosphoryl-transfer/EI component)
PP_0794	***fruK***	**Q88PQ4**	**Phosphofructokinase (1-PFK) — this study**
PP_0795	*fruA*	Q88PQ3	Protein-Nπ-phosphohistidine–D-fructose phosphotransferase (fructose PTS EII permease)
PP_0796	–	Q88PQ2	Cytoplasmic protein

The co-localization of *fruK* between *fruB* (PTS phosphotransfer) and *fruA* (fructose EII permease), with the *cra* regulator immediately upstream, independently corroborates FruK's role in the fructose-PTS utilization module and its Cra/F1P control (see §5.3). Conserved gene neighborhood is itself strong evidence of shared function and co-regulation.

3. Primary Function: The Catalyzed Reaction and Substrate Specificity

FruK is an **ATP-dependent sugar-phosphate kinase** that phosphorylates the C6 hydroxyl of **fructose-1-phosphate**, generating **fructose-1,6-bisphosphate** (UniProt Q88PQ4; Rhea:14213; EC 2.7.1.56). Associated molecular functions annotated for the protein are **1-phosphofructokinase activity** (GO:0008662) and **ATP binding** (GO:0005524).

Substrate specificity. The physiological substrate is the fructose-PTS product fructose-1-phosphate — *not* fructose-6-phosphate. This distinction is functionally critical: the fructose PTS phosphorylates the incoming sugar at C1, so a specialized 1-phosphofructokinase (rather than the glycolytic 6-phosphofructokinase) is required to complete phosphorylation to the bis-phosphate. Biochemical characterization of orthologous FruK enzymes confirms this reaction directly:

- In *Aeromonas hydrophila*, "1-Phosphofructokinase (1-PFK), which converts the product of the PTS reaction to fructose 1,6-diphosphate, was present … grown with fructose" (Binet et al. 1998, P 9579084).
- In *Xanthomonas campestris*, "The fructose 1-phosphate produced by the phosphotransferase system is phosphorylated into fructose 1,6-bisphosphate by a 1-phosphofructokinase" (de Crécy-Lagard et al. 1991, P 1655739).

Structure/sequence-based evidence of catalytic competence. The 315-aa sequence carries a single Carbohydrate-kinase PfkB domain (UniProt feature, residues 12–290) and displays the two diagnostic ribokinase-superfamily signatures: (i) an N-terminal glycine-rich loop `…GGFLGGDN…` (GG at positions 58 and 62), and (ii) the C-terminal catalytic motif **`GAGD`** **(residues 251–254**, within `…VASTVGAGDSLVAG…`). In the ribokinase/PfkB fold, the aspartate of the GAGD motif is the conserved catalytic base that deprotonates the sugar hydroxyl to permit in-line phosphoryl transfer from ATP. The presence and spacing of these motifs provide sequence-level evidence — independent of database text annotation — that FruK is a bona fide, catalytically competent PfkB-family sugar kinase.

Family / structural basis. FruK belongs to the **PfkB family** of carbohydrate kinases, which share a **ribokinase-like α/β/α fold** and include *E. coli* PfkB, ribokinase, and phosphotagatokinase (Wu et al. 1991,

).

P 1850730

The family is evolutionarily and mechanistically distinct from the major glycolytic PfkA-type 6-phosphofructokinase: the *E. coli* FruK sequence shows "little similarity to the major 6-phosphofructokinase (pfkA) ... but there is 27% ... identity ... with the minor 6-phosphofructokinase (pfkB)" (Orchard & Kornberg 1990,

).

P 1981619

The five InterPro signatures on Q88PQ4 (PfkB domain, ribokinase-like, PfkB conserved site, tagatose/fructose PfkB-type kinase) place the *P. putida* protein squarely in this family.

Quaternary structure (inference). No experimental oligomeric-state data exist for *P. putida* FruK; characterized PfkB-family 1-phosphofructokinases (e.g., *E. coli* FruK) are typically homo-oligomers (commonly dimeric), and *E. coli* FruK has additionally been shown to form higher-order hetero-complexes with the Cra regulator (Weeramange et al. 2024,

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P 38723750

The oligomeric state of the *P. putida* enzyme remains to be determined experimentally.

4. Subcellular Localization

FruK carries out its function in the **cytoplasm (cytosol)**. UniProt assigns the cytosol (GO:0005829) by phylogenetic inference. Consistent with this, the 315-aa sequence contains no signal peptide or transmembrane segment, and PfkB-family sugar kinases are canonically soluble cytoplasmic enzymes. Mechanistically the localization is expected: the membrane-embedded fructose PTS (FruA/FruB) delivers fructose-1-phosphate into the cytoplasm, where soluble FruK phosphorylates it using cytoplasmic ATP. *(Evidence class: inference from sequence, family, and biochemical logic — no direct experimental localization study specific to P. putida FruK was found.)*

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5. Biochemical Pathway and Physiological Role

5.1 Fructose utilization pathway (the "fru" system)

P. putida KT2440 has an unusually minimal PTS: the genome encodes only five PTS proteins, of which **FruA and FruB constitute the complete fructose intake system** (Pflüger & de Lorenzo 2008,

P 18296519

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The pathway operates as:

1. **Import + phosphorylation:** fructose → **fructose-1-phosphate (F1P)** by the fructose PTS (PEP-dependent; FruB relays phosphate to the FruA EII permease).
2. **FruK step: F1P + ATP → fructose-1,6-bisphosphate** (this enzyme).
3. **Aldol cleavage:** fructose-1,6-bisphosphate → dihydroxyacetone-P + glyceraldehyde-3-P (fructose-bisphosphate aldolase), feeding lower central metabolism.

5.2 Why FruK is obligatory in *P. putida*

P. putida catabolizes hexoses **exclusively through the Entner–Doudoroff (ED) pathway "due to the absence of 6-phosphofructokinase"** (Chavarría et al. 2013,

P 23301697

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There is therefore **no glycolytic F6P→FBP reaction**; forcing the Embden–Meyerhof–Parnas route by heterologous *E. coli* pfkA was even detrimental to growth and oxidative-stress tolerance. Consequently, FruK's 1-PFK reaction is the *only* route by which PTS-imported fructose is converted to a bisphosphate that aldolase can cleave — making FruK indispensable for fructose catabolism. The essentiality of FruK for fructose growth is directly demonstrated in *E. coli*, where "The fruK mutants were unable to utilize … fructose and fructose-1-phosphate" (Molchanova et al. 1992,

P 1332910

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5.3 Regulatory role — a feed-forward circuit

FruK is embedded in a tight regulatory loop. The **fruBKA operon is repressed by the Cra (FruR) transcription factor**, and the *sole physiological effector* that relieves this repression in *P. putida* is **fructose-1-phosphate — FruK's own substrate**: "Cra(PP) represses expression in vivo of the cognate fruBKA operon in a fashion dependent just on F1P, ruling out any other physiological effector" (Chavarría et al. 2014,

P 24918052

).

Thus arriving fructose (as F1P) both (a) feeds FruK catalysis and (b) induces transcription of *fruK* itself, giving feed-forward activation of the pathway. In the wider gammaproteobacterial context, "transcription of the fru operon … and the 1-phosphofructokinase FruK is repressed by FruR in the absence of the inducer F1P" (Yoon et al. 2021,

P 33476373

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5.4 Possible moonlighting / broader connections (lower confidence, from orthologs)

- In *E. coli*, FruK can run its **reverse reaction** (FBP→F1P) at physiological concentrations and **binds the Cra regulator directly with nanomolar affinity**, adding a protein-level layer to central-metabolism regulation; the authors note these findings "might have wide-spread relevance to other γ-proteobacteria, which conserve both Cra and FruK" (Weeramange et al. 2024,

  P 38723750

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  Whether *P. putida* FruK moonlights similarly is untested.
- The *P. putida* fructose PTS **cross-talks with the nitrogen-related PTS branch** ($EIIA^{Ntr}$), linking carbon and nitrogen sensing (Pflüger & de Lorenzo 2008,

  P 18296519

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  This is upstream of FruK but defines the regulatory neighbourhood in which the enzyme operates.

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6. Supported and Refuted Hypotheses

Supported: - H1 — FruK is a 1-phosphofructokinase (EC 2.7.1.56) converting F1P + ATP → fructose-1,6-bisphosphate. **Strongly supported** (UniProt/Rhea annotation; ortholog biochemistry PMIDs 9579084, 1655739). - H2 — FruK belongs to the PfkB/ribokinase-like family, distinct from glycolytic PfkA. **Strongly supported** (InterPro/Pfam; PMIDs 1850730, 1981619). - H3 — FruK acts in the fructose-PTS (fruBKA) catabolic pathway, downstream of FruA/FruB.

Supported (PMIDs 33476373, 18296519, 24918052). - H4 — FruK is functionally obligatory for fructose use because *P. putida* lacks 6-PFK and uses the ED pathway. **Supported** (P 23301697; ortholog essentiality P 1332910). - H5 — FruK's substrate F1P is the physiological inducer of its own operon via Cra. **Supported** (P 24918052). - H6 — FruK is a soluble cytoplasmic enzyme. **Supported by inference** (UniProt GO cytosol; no signal/TM; family logic). - H7 — Genomic context: *fruK* lies in the fruBKA operon with *cra* upstream, and its sequence carries the PfkB catalytic motifs. **Supported by bioinformatics** (locus map PP_0792 *cra* – PP_0793 *fruB* – PP_0794 *fruK* – PP_0795 *fruA*; GAGD catalytic-Asp motif + Gly-rich loop).

Refuted / excluded: - FruK is **not** a glycolytic 6-phosphofructokinase and does **not** act on fructose-6-phosphate (excluded by substrate specificity and family divergence from PfkA). - The gene is **not** a mis-annotation of a different same-symbol gene: organism-specific literature confirms *P. putida* FruK identity.

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7. Limitations and Future Directions

- **No enzyme-specific experimental study of *P. putida* Q88PQ4 itself** (kinetics, Km/kcat, structure) was located; the catalytic assignment rests on high-quality UniProt/Rhea annotation plus biochemistry of close orthologs (*E. coli*, *Xanthomonas*, *Aeromonas*, *Rhodobacter*).
- **Localization is inferred**, not experimentally demonstrated for this protein.
- **Possible moonlighting** (direct Cra binding, reverse FBP→F1P reaction) is documented only in *E. coli* FruK (P 38723750) and remains to be tested in *P. putida*.
- Future work: recombinant purification and steady-state kinetics of PP_0794; a crystal/AlphaFold-guided active-site analysis of the ribokinase-like fold; a ΔfruK strain growth/flux

study on fructose vs. glucose; and testing whether *P. putida* FruK physically interacts with Cra.

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8. Key References

- Yoon et al. 2021, *J Biol Chem/J Bacteriol* —

  P 33476373

  (P. putida fru operon; FruK = 1-PFK; F1P inducer).
- Chavarría et al. 2014 —

  P 24918052

  (F1P is the sole Cra effector; fruBKA operon).
- Chavarría et al. 2013, *Environ Microbiol* —

  P 23301697

  (P. putida lacks 6-PFK; ED pathway exclusive).
- Pflüger & de Lorenzo 2008, *J Bacteriol* —

  P 18296519

  (FruA/FruB = complete fructose PTS; C/N cross-talk).
- Weeramange et al. 2024 —

  P 38723750

  (E. coli FruK reverse reaction & Cra binding; relevance to γ-proteobacteria).
- Wu et al. 1991 —

  P 1850730

  (PfkB family definition).
- Orchard & Kornberg 1990 —

  P 1981619

  (fruK vs pfkA/pfkB; fruFKA operon).

- de Crécy-Lagard et al. 1991 —

  P 1655739

  (Xanthomonas FruK biochemistry).
- Binet et al. 1998 —

  P 9579084

  (Aeromonas 1-PFK on PTS product).
- Molchanova et al. 1992 —

  P 1332910

  (E. coli fruK essential for fructose growth).
